

Smithsonian Fossil Palynomorph Detection and Classification

Initial Report — Fall 2026

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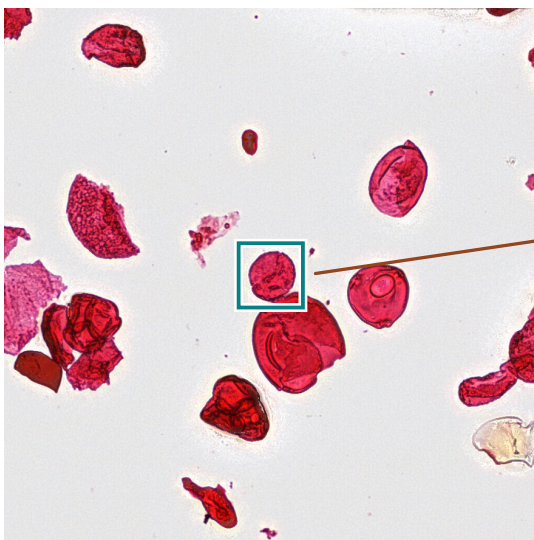
Applied Machine Learning & Data Science Projects
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Faculty Mentor: Dr. Arko Barman, Rice University

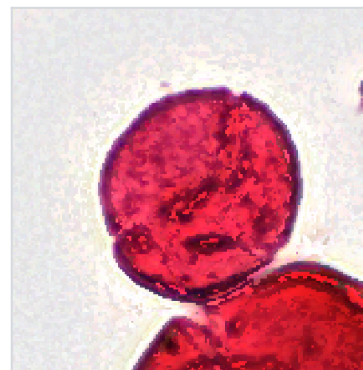
PhD Mentor: Nhi Le, Rice University

Sponsors: Ingrid Romero and Scott Wing, Smithsonian Institution

A A crowded microscope image



B Inspect the specimen



Example expert label

Bombacacidites sp.

DETECTION Where is the specimen?

CLASSIFICATION What type is it?

Fig. 1. From detection to identification. A focus-stacked microscope image contains multiple palynomorphs and surrounding material (left). The highlighted specimen is enlarged for category identification (right). The box and category label are derived from expert annotations and illustrate the task rather than model predictions.